

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
QUADRATIC EQUATIONS | LEVEL 2*Lengthy – correct method, sustained computation.*

1. If $\frac{3+2\sqrt{2}}{3-\sqrt{2}} = a + b\sqrt{2}$, where $a, b \in \mathbb{Q}$, then a and b are respectively
- (A) $\frac{9}{7}, \frac{13}{7}$ (B) $\frac{13}{7}, \frac{9}{7}$
- (C) $\frac{13}{7}, \frac{7}{9}$ (D) $\frac{7}{9}, \frac{7}{13}$
2. The number of solutions of the equation $\log(-2x) = 2\log(x+1)$ is
- (A) 2 (B) 1
- (C) 0 (D) 3
3. The sum of the solutions of the equation $9^x - 6 \cdot 3^x + 8 = 0$ is
- (A) $\log_3 4$ (B) $\log_3 6$
- (C) $\log_3 8$ (D) $\log_3 2$
4. Let $p, q \in \{1, 2, 3, 4\}$. The number of equations of the form $px^2 + qx + 1 = 0$ having real roots is
- (A) 15 (B) 8
- (C) 9 (D) 7
5. If $b \in \mathbb{R}^+$, then the roots of $(2+b)x^2 + (3+b)x + (4+b) = 0$ are
- (A) cannot be predicted (B) real and distinct
- (C) imaginary (D) real and equal
6. If the roots of $ax^2 + x + b = 0$ are real and different, then the roots of $x^2 - 4\sqrt{ab}x + 1 = 0$ are
- (A) real (B) rational
- (C) imaginary (D) irrational
7. If $a < c < b$, then the roots of $(a-b)^2x^2 + 2(a+b-2c)x + 1 = 0$ are
- (A) equal
- (B) real
- (C) one real and one imaginary

(D) imaginary

8. If α, β are the roots of $Ax^2 + Bx + C = 0$ and α^2, β^2 are the roots of $x^2 + px + q = 0$, then p equals

(A) $\frac{2AC - B^2}{A^2}$

(B) $\frac{4AC - B^2}{A^2}$

(C) $\frac{B^2 - 4AC}{A^2}$

(D) none of these

9. If exactly one root of the quadratic equation $f(x) = ax^2 + bx + c = 0$ is at infinity, then

(A) both (A) and (C)

(B) b tends to zero

(C) a tends to zero

(D) b must not be zero

10. The value of m for which $\frac{a}{x+a+m} + \frac{b}{x+b+m} = 1$ has roots equal in magnitude and opposite in sign is

(A) $\frac{a+b}{a-b}$

(B) $\frac{a-b}{a+b}$

(C) -1

(D) 0

11. The value of a for which $x^7 + ax^2 + 3 = 0$ and $x^8 + ax^3 + 3 = 0$ have a common root is

(A) -3

(B) -4

(C) -2

(D) 1

12. If $x^2 + 3x + 3 = 0$ and $ax^2 + bx + 1 = 0$, $a, b \in \mathbb{Q}$, have a common root, then $3a + b$ equals

(A) 2

(B) $\frac{1}{3}$

(C) 4

(D) 1

13. If α, β are the roots of $x^2 - px + r = 0$ and $\alpha + 1, \beta - 1$ are the roots of $x^2 - qx + r = 0$, then r is

(A) $\frac{p^2 - 1}{4}$

(B) $\frac{p - 1}{4}$

(C) $\frac{q^2 + 1}{4}$

(D) $\frac{q + 1}{4}$

14. The sum of all values of p for which the vertex of the parabola $y = x^2 + 2px + 13$ lies at a distance 5 from the origin is

(A) 7

(B) 8

(C) 6

(D) 0

15. If α, β are the roots of $2x^2 + 4x - 5 = 0$, then the equation whose roots are $2\alpha - 3$ and $2\beta - 3$ is
- (A) $x^2 + 10x - 11 = 0$ (B) $11x^2 + 10x - 1 = 0$
- (C) $x^2 + 10x + 11 = 0$ (D) $11x^2 - 10x + 1 = 0$